

Influence Parameter Calibration of UBI Equilibrium

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Background

δ describes how agent i perceives agent j 's response to i 's labor choices

- ▶ $\delta > 0$: greater good - UBI has a positive effect on agents, leading to improved labor participation rates and welfare
- ▶ $\delta < 0$: freeloader - UBI has a negative effect on agents, reducing labor participation rates and overall welfare

Goal: Use real-world data to estimate δ in actual economies

Parameters

Parameter	Meaning
α_i	risk aversion
$w_{i,t}$	wage rates
I	population
$L_{i,t}$	labor proportion
β_i	labor preference
γ_i	leisure preference
$\lambda_{keep}, \lambda_{ubi}$	UBI proportions

We also note the utility function

$$U(t, c, l) = -l^\beta (1 - l)^\gamma e^{-\rho t - \alpha c}$$

Formula for δ

Starting from Definition 3.1 in the paper:

$$\alpha_i w_{i,t} \left(\lambda_{keep} + \frac{\lambda_{ubi}}{I} (1 + \delta(I - 1)) \right) = \frac{u'_i(L_{i,t})}{u_i(L_{i,t})}$$

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$$\alpha_i w_{i,t} \left(\lambda_{keep} + \frac{\lambda_{ubi}}{I} (1 + \delta(I - 1)) \right) = \frac{\beta_i}{L_{i,t}} - \frac{\gamma_i}{1 - L_{i,t}}$$

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$$\vdots$$

$$\delta = \frac{I}{I - 1} \left(\frac{1}{\alpha_i w_{i,t} \lambda_{ubi}} \left(\frac{\beta_i}{L_{i,t}} - \frac{\gamma_i}{1 - L_{i,t}} \right) - \frac{\lambda_{keep} + \frac{\lambda_{ubi}}{I}}{\lambda_{ubi}} \right)$$

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$$\lim_{I \rightarrow \infty} \delta = \frac{1}{\alpha w \lambda_{ubi}} \left(\frac{\beta}{L} - \frac{\gamma}{1 - L} \right) - \frac{\lambda_{keep}}{\lambda_{ubi}}$$

Estimating Risk Aversion α

Methodology: Assuming a risk aversion function $\alpha = r(W) = \frac{c}{W^h}$, replicated Szpiro's paper using insurance claims and wealth data to empirically estimate c for an economy.

The utility maximizing number of assets insured is modeled as, $I = W - \frac{\lambda}{r(W)}$ and the claims rate as $Q = Iq$, combining you get that

$$Q = qW - \frac{q\lambda}{r(W)}$$

Estimating Risk Aversion α

Plugging in the risk aversion function $r(W) = \frac{c}{W^h}$:

$$Q_t = qW_t - \frac{q}{c}\lambda_t W_t^h.$$

We estimate the following regression:

$$Q_t = bW_t + n(\lambda_t W_t^h),$$

where $b = q$, $n = -\frac{q}{c}$. Rearranging and solving for c , we get:

$$n = -\frac{b}{c} \Rightarrow c = -\frac{b}{n}.$$

Main challenges: OECD Wealth data was incomplete and Claims data was noisy leading to unreasonable estimates for some economies.

Other Parameters

Labor proportion

$$L = \frac{\text{annual hours of labor}}{\text{annual waking hours}}$$

Wage rates

$$w = \frac{\text{annual earnings}}{L} = \frac{\text{gross labor costs}}{L}$$

We simply guess the values of $\beta, \gamma, \lambda_{keep}$ and λ_{ubi} , but we can say

$$\lambda_{keep} + \lambda_{ubi} = 1 - \text{tax wedge}$$

Guessing β, γ

$$\delta = \frac{1}{\alpha w \lambda_{ubi}} \left(\frac{\beta}{L} - \frac{\gamma}{1-L} \right) - \frac{\lambda_{keep}}{\lambda_{ubi}}$$

For $\delta > 0$, we at the very least need

$$\frac{\beta}{L} - \frac{\gamma}{1-L} > 0 \quad \Rightarrow \quad \frac{\gamma}{\beta} > \frac{1-L}{L}$$

Try Sweden 2024 data: 1450 hours/year $\Rightarrow L \approx 0.25$

$$\frac{\gamma}{\beta} > \frac{1-L}{L} \approx \frac{0.75}{0.25} = 3 \quad \Rightarrow \quad \beta = -0.2, \gamma = -0.8$$

Results

There are none.

Results

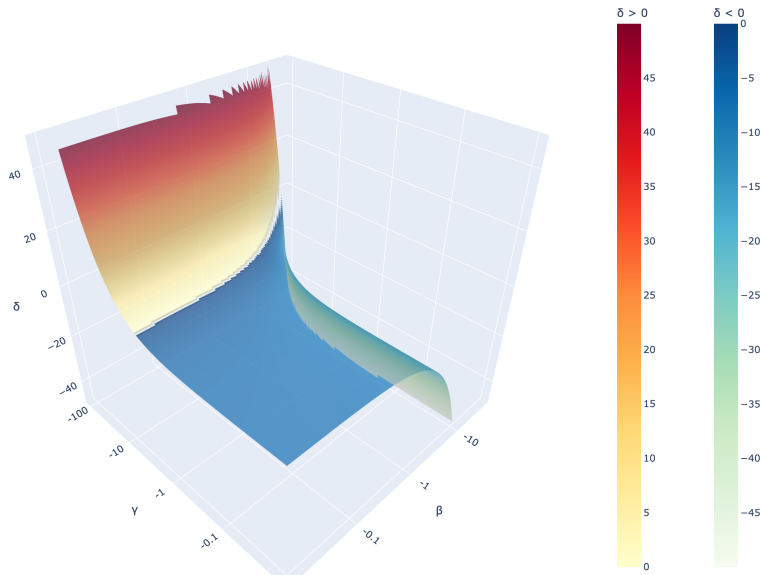
$$\beta = -0.2 \quad \gamma = -0.8$$

$$\lambda_{keep} = 0.5 \quad \lambda_{ubi} = 0.1$$

	2019	2020	2021	2022
Sweden	-4.807295	-4.834609	-4.812561	-4.802891
United States	-4.229834	-4.305471	-4.223981	-4.202037

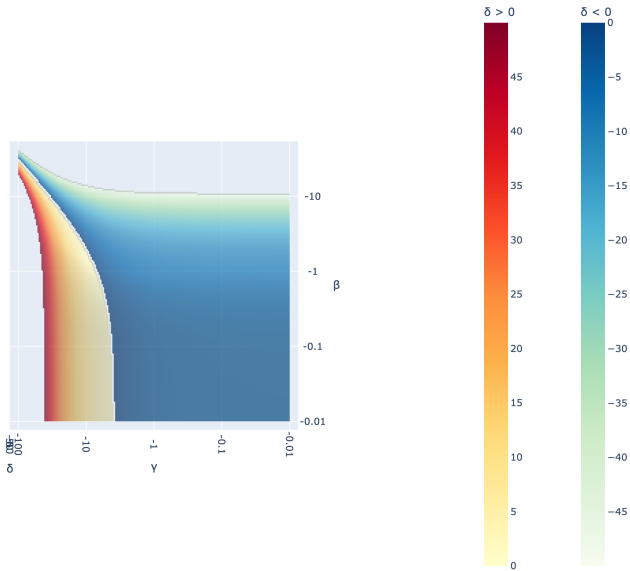
δ for Sweden in 2024

$\lambda_{\text{keep}}=0.5$, $\lambda_{\text{ubi}}=0.1$, $\sigma=3.04\text{e-}05$



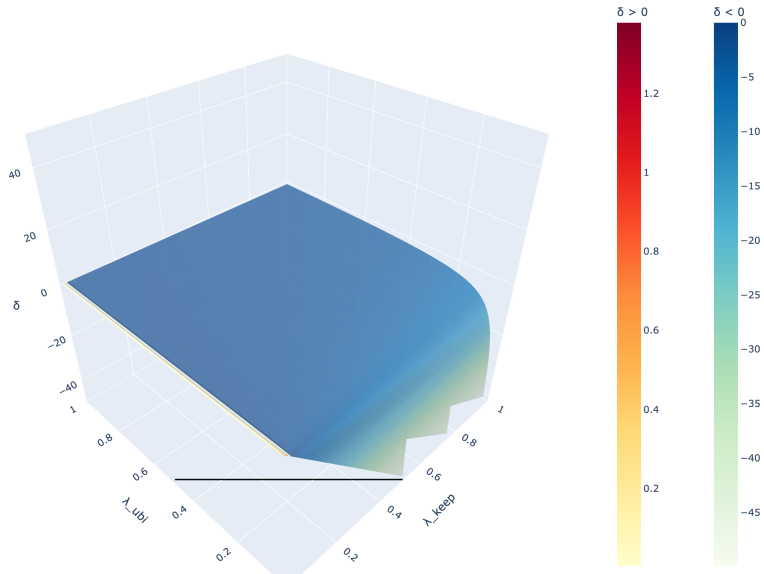
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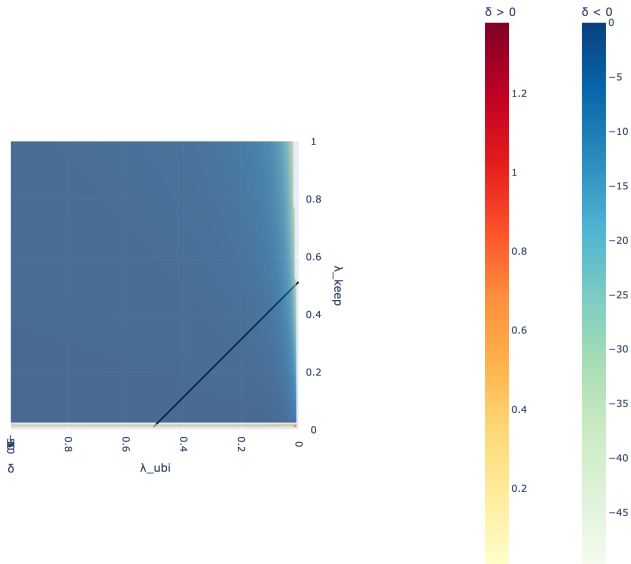
δ for Sweden in 2024

$\beta=-0.2$, $\delta=-0.8$, $\alpha=3.04e-05$



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Why don't our results agree with UBI experiments?

But haven't past UBI experiments found that workers continue to work even when receiving UBI?

Issues on both sides:

- ▶ UBI experiments have been very small scale
 - ▶ Typically conducted among low-income workers
- ▶ Assumed homogeneous agents and time invariance
 - ▶ same wages and taxes
 - ▶ same labor proportions
 - ▶ none of the above changes over time

Next steps: Model income brackets while maintaining large population size

Thank you!